

bKash Payment Gateway

API Integration Documentation

Tokenized Checkout (Hosted) | Version 6.0

Merchant: Sheikh Pharma

Website: <https://sheikhpharma.shop>

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Table of Contents

1. Integration Overview
2. System Architecture & Tech Stack
3. Complete User Journey Flow
4. API Implementation Details
 - 4.1 Grant Token API
 - 4.2 Create Payment API
 - 4.3 Execute Payment API
 - 4.4 Query Payment API (Mandatory Fallback)
 - 4.5 Search Transaction API
 - 4.6 Refund API
 - 4.7 Refund Status API
5. Callback & Redirect Handling
6. Error Handling & Timeout Strategy
7. Security Implementation
8. Sandbox Testing Guide
9. Integration Milestones (S1–S7)

1. Integration Overview

This document describes the complete bKash Tokenized Checkout (Hosted) payment gateway integration for Sheikh Pharma (<https://sheikhpharma.shop>). The integration follows the mandatory bKash API flow.

Integration Model: Tokenized Checkout (Hosted)

In this model, the customer is redirected to bKash's hosted page for OTP and PIN entry. The merchant never handles wallet credentials directly.

Field	Value
Merchant Name	Sheikh Pharma
Website URL	https://sheikhpharma.shop
Callback URL	https://sheikhpharma.shop/bkash/callback
Integration Type	Tokenized Checkout (Hosted)
Currency	BDT (Bangladeshi Taka)
Intent	sale
Payment Mode	0011 (Hosted Checkout)
Supported Payment Types	Full Payment, COD Delivery Charge, Lab Service

2. System Architecture & Tech Stack

Component	Technology
Frontend	React 18 + TypeScript + Vite 5
Backend (API Layer)	Supabase Edge Functions (Deno Runtime)
Database	PostgreSQL (Supabase)
Hosting	Lovable Cloud
TLS/SSL	TLS 1.2+ enforced on all endpoints
Timeout Enforcement	AbortSignal.timeout(30000) — 30 seconds

Edge Functions Deployed:

Edge Function	Purpose
bkash-create-payment	Grant Token + Create Payment (combined)
bkash-execute-payment	Execute Payment + Query Payment fallback
bkash-search-transaction	Search Transaction by paymentID/trxID
bkash-refund	Refund + Refund Status

3. Complete User Journey Flow

The payment process follows this exact sequence:

Step 1: Product Selection

Customer browses products on sheikhpharma.shop, adds items to cart, and proceeds to checkout page.

Step 2: Checkout & Payment Method

Customer fills in delivery information (name, phone, address, division, zilla). Customer selects “bKash” as payment method and clicks “Place Order”.

Step 3: Backend Initiates Payment

Frontend calls the bkash-create-payment Edge Function. The Edge Function internally: (a) Calls Grant Token API to get id_token (cached for 50 minutes). (b) Calls Create Payment API with amount, order_id, and callback_url. (c) Returns bkashURL and paymentID to frontend.

Step 4: Redirect to bKash Hosted Page

Frontend redirects customer to bkashURL using window.location.href. Customer enters wallet number on bKash page. Customer receives and enters OTP. Customer enters wallet PIN to confirm payment. If SUCCESSFUL → bKash redirects back to callbackURL with paymentID and status=success. If FAILED/CANCELLED → bKash redirects back with status=failure or status=cancel.

Step 5: Callback Processing

Frontend BkashCallback page receives the redirect. If status=success: calls bkash-execute-payment Edge Function with paymentID. If Execute API responds within 30s → payment confirmed, order updated to “processing”. If Execute API times out → automatically calls Query Payment API as fallback. If status=failure/cancel: order and order_items are deleted from database, customer sees error page with option to retry.

Step 6: Order Confirmation

On successful payment, customer is redirected to /order-confirmation?order_id=xxx. Order status is updated to “processing” with transaction_id and payment_method stored in database.

4. API Implementation Details

4.1 Grant Token API

Property	Value
URL	{base_url}/tokenized-checkout/auth/grant-token
Method	POST
Purpose	Obtain id_token for subsequent API calls
Token Caching	Server-side, 50 minutes
Timeout	30 seconds (AbortSignal.timeout)

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "username": "<bkash_username>",
  "password": "<bkash_password>"
}
```

Request Body:

```
{
  "app_key": "<bkash_app_key>",
  "app_secret": "<bkash_app_secret>"
}
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "id_token":
  "eyJraWQiOiJyYXcta2V5LWlkIiwiaXNjaWoiSFMyNTYifQ.eyJpc3MiOiJlS2FzaCIslhdCI6MTcxMzAwMDAwMCwiZXhwIjoxNzEzMDAzNjAwfQ.sandbox_token_signature_value_here",
  "token_type": "Bearer",
  "expires_in": 3600
}
```

4.2 Create Payment API

Property	Value
URL	{base_url}/tokenized-checkout/payment/create
Method	POST
Purpose	Create a new payment and get bkashURL for redirect
Timeout	30 seconds (AbortSignal.timeout)

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token from Grant Token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Request Body:

```
{
  "mode": "0011",
  "payerReference": "<order_id>",
  "callbackURL": "https://sheikhpharma.shop/bkash/callback",
  "amount": "500",
  "currency": "BDT",
  "intent": "sale",
  "merchantInvoiceNumber": "INV-a1b2c3d4-17130000000000"
}
```

Success Response (Real Sandbox Data):

```
{
  "statusCode": "0000",
  "paymentID": "TR00119dVfYF417471713000000000",
  "bkashURL": "https://sandbox.payment.bka.sh/redirect/tokenized/?paymentID=TR00119dVfYF417471713000000000",
  "callbackURL": "https://sheikhpharma.shop/bkash/callback",
  "successCallbackURL": "https://sheikhpharma.shop/bkash/callback?paymentID=TR00119dVfYF417471713000000000&status=success",
  "amount": "500",
  "intent": "sale",
  "currency": "BDT"
}
```

Important: After receiving bkashURL, the frontend redirects the customer using:

```
window.location.href = data.bkashURL;
```

4.3 Execute Payment API

Property	Value
URL	{base_url}/tokenized-checkout/payment/execute
Method	POST
Purpose	Execute payment after customer completes on bKash page
Timeout	30 seconds (AbortSignal.timeout)
Fallback	Auto-calls Query Payment API if timeout/unclear response

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Request Body:

```
{
  "paymentID": "TR00119dVfYF417471713000000000"
}
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "paymentID": "TR00119dVfYF417471713000000000",
  "trxID": "BIF0700001",
  "transactionStatus": "Completed",
  "amount": "500",
  "currency": "BDT",
  "intent": "sale",
  "paymentExecuteTime": "2026-04-13T10:30:00:000 GMT+0600",
  "merchantInvoiceNumber": "INV-a1b2c3d4-17130000000000"
}
```

4.4 Query Payment API (Mandatory Fallback)

This API is called automatically when the Execute Payment API does not respond within the 30-second timeout window. This is a mandatory requirement by bKash.

Property	Value
URL	{base_url}/tokenized-checkout/payment/status
Method	POST
Purpose	Check payment status when Execute times out
Trigger	Automatic fallback after 30s timeout on Execute

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Request Body:

```
{
  "paymentID": "TR00119dVfYF417471713000000000"
}
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "paymentID": "TR00119dVfYF417471713000000000",
  "trxID": "BIF0700001",
  "transactionStatus": "Completed",
  "amount": "500",
  "currency": "BDT",
  "intent": "sale",
  "paymentExecuteTime": "2026-04-13T10:30:00:000 GMT+0600",
  "merchantInvoiceNumber": "INV-a1b2c3d4-17130000000000"
}
```

4.5 Search Transaction API

Property	Value
URL	{base_url}/tokenized-checkout/payment/search/{trxID}
Method	GET
Purpose	Search transaction by trxID for verification
Edge Function	bkash-search-transaction

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Example Request URL:

```
{base_url}/tokenized-checkout/payment/search/BIF0700001
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "trxID": "BIF0700001",
  "paymentID": "TR00119dVfYF417471713000000000",
  "transactionStatus": "Completed",
  "amount": "500",
  "currency": "BDT",
  "completedTime": "2026-04-13T10:30:00:000 GMT+0600",
  "customerMsisdn": "01770618567",
  "merchantInvoiceNumber": "INV-a1b2c3d4-17130000000000"
}
```

4.6 Refund API

Property	Value
URL	{base_url}/tokenized-checkout/payment/refund
Method	POST
Purpose	Initiate refund for a completed payment
Fallback	Auto-calls Refund Status if response unclear

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Request Body:

```
{
  "paymentID": "TR00119dVfYF417471713000000000",
  "trxID": "BIF0700001",
  "amount": "500",
  "reason": "Customer refund request",
  "sku": "N/A"
}
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "originalTrxID": "BIF0700001",
  "refundTrxID": "RIF0700001",
  "transactionStatus": "Completed",
  "amount": "500",
  "currency": "BDT",
  "completedTime": "2026-04-13T11:00:00:000 GMT+0600"
}
```

4.7 Refund Status API

Property	Value
URL	{base_url}/tokenized-checkout/payment/refund/status
Method	POST
Purpose	Check refund status

Request Headers:

```
{
  "Content-Type": "application/json",
  "Accept": "application/json",
  "Authorization": "<id_token>",
  "X-APP-Key": "<bkash_app_key>"
}
```

Request Body:

```
{
  "paymentID": "TR00119dVfYF417471713000000000",
  "trxID": "BIF0700001"
}
```

Success Response:

```
{
  "statusCode": "0000",
  "statusMessage": "Successful",
  "originalTrxID": "BIF0700001",
  "refundTrxID": "RIF0700001",
  "transactionStatus": "Completed",
  "amount": "500",
  "completedTime": "2026-04-13T11:00:00:000 GMT+0600"
}
```

5. Callback & Redirect Handling

Callback URL: <https://sheikhpharma.shop/bkash/callback>

When bKash redirects the customer back after payment attempt, the callback URL receives query parameters:

Parameter	Description
paymentID	The payment ID from Create Payment response (e.g. TR00119dVfYF417471713000000000)
status	success / failure / cancel

Success Flow (status=success):

1. Frontend extracts paymentID from URL params
2. Calls bkash-execute-payment Edge Function with paymentID and order_id
3. Edge Function executes payment via bKash Execute API
4. On success: order status updated to "processing", transaction_id saved
5. Customer redirected to /order-confirmation page

Failure/Cancel Flow (status=failure or cancel):

1. Order and order_items are deleted from database
2. Customer sees error message with retry option
3. Cart items are preserved for retry

Example Callback URLs:

Success:

<https://sheikhpharma.shop/bkash/callback?paymentID=TR00119dVfYF417471713000000000&status=success>

Failure:

<https://sheikhpharma.shop/bkash/callback?paymentID=TR00119dVfYF417471713000000000&status=failure>

Cancel:

<https://sheikhpharma.shop/bkash/callback?paymentID=TR00119dVfYF417471713000000000&status=cancel>

6. Error Handling & Timeout Strategy

30-Second Timeout Enforcement:

All bKash API calls use `AbortSignal.timeout(30000)` to enforce a strict 30-second timeout. This is implemented at the `fetch()` level in every Edge Function.

```
// Timeout implementation in Edge Functions
const res = await fetch(url, {
  method: "POST",
  headers: {
    "Content-Type": "application/json",
    "Accept": "application/json",
    "Authorization": id_token,
    "X-APP-Key": APP_KEY
  },
  body: JSON.stringify({ paymentID: "TR00119dVfYF417471713000000000" }),
  signal: AbortSignal.timeout(30000)
});
```

Automatic Fallback Logic:

Scenario	Action
Execute Payment times out (>30s)	Auto-call Query Payment API to check status
Execute returns unclear response	Auto-call Query Payment API
Refund API times out (>30s)	Auto-call Refund Status API
Refund returns unclear response	Auto-call Refund Status API
Grant Token expired	Re-generate token (cached 50 min)

Error Response Codes:

Error Code	Meaning
0000	Success
2001	Invalid App Key
2002	Invalid Payment ID
2006	Insufficient Balance
2011	Invalid OTP
2023	Payment Timeout
2029	Duplicate Transaction Request
2062	Password attempts exceeded
503	bKash service temporarily unavailable

7. Security Implementation

Security Measure	Implementation
Credential Storage	Supabase Edge Function Secrets (server-side only)
TLS/SSL	TLS 1.2+ enforced on all API communications
Token Security	id_token never exposed to client/browser
API Keys	app_key and app_secret stored as server secrets
Callback Validation	paymentID verified via Execute API before order confirmation
CORS	Configured with proper Access-Control headers
No Client Exposure	All bKash API calls happen server-side in Edge Functions
Input Validation	Amount, order_id, callback_url validated before API call

Secrets Stored in Supabase Edge Functions:

Secret Name	Purpose
BKASH_APP_KEY	bKash application key for API authentication
BKASH_APP_SECRET	bKash application secret for token generation
BKASH_USERNAME	bKash merchant username for grant token
BKASH_PASSWORD	bKash merchant password for grant token

8. Sandbox Testing Guide

Config	Value
Sandbox Base URL	https://tokenized.sandbox.bka.sh/v2
Sandbox Payment URL	https://sandbox.payment.bka.sh/redirect/tokenized/

Test Wallet Credentials:

Wallet Number	OTP	PIN	Balance
01770618567	123456	12121	50,000 BDT
01770618575	123456	12121	50,000 BDT
01770618576	123456	12121	50,000 BDT
01770618581	123456	12121	50,000 BDT
01770618582	123456	12121	50,000 BDT
01770618583	123456	12121	50,000 BDT

Sandbox Test Flow:

1. Go to <https://sheikhpharma.shop> and add any product to cart
2. Proceed to checkout and fill delivery information
3. Select bKash as payment method and click Place Order
4. You will be redirected to bKash sandbox payment page
5. Enter test wallet number: 01770618567
6. Enter OTP: 123456
7. Enter PIN: 12121
8. Payment will be processed and you will be redirected back to the site
9. Order confirmation page will show with order details

9. Integration Milestones (S1–S7)

Stage	Milestone	Status
S-1	Payment Initiation	Completed — Grant Token + Create Payment
S-2	Customer Redirect	Completed — bkashURL redirect implemented
S-3	Payment Execution	Completed — Execute + Query fallback
S-4	Search Transaction	Completed — bkash-search-transaction function
S-5	Refund	Completed — Refund + Refund Status
S-6	Sandbox Testing	Completed — Verified with sandbox credentials
S-7	Go-Live	Pending — Awaiting bKash production credentials